CalmSense

1. Project Overview

CalmSense is an end-to-end multimodal stress detection system. It processes raw biosignals from wearable sensors, extracts 60+ clinically grounded biomarkers, and classifies physiological states (baseline, stress, amusement) using both classical machine learning and deep learning. All models use Leave-One-Subject-Out cross-validation for subject-independent evaluation. Predictions are explained via SHAP, LIME, and clinical biomarker mapping. The system is deployed through a FastAPI backend and React dashboard with Docker support.

Aspect	**Details**
Dataset	WESAD — 15 subjects, chest-worn RespiBAN, 4 experimental conditions
Signals	ECG, EDA, Temperature, Accelerometer, Respiration (all at 700 Hz)
Classification	3-class: Baseline, Stress (TSST), Amusement
Features	60+ across 7 domains (HRV time/freq/nonlinear, EDA, temp, resp, accel)
ML Models	LR, SVM, Random Forest, XGBoost, LightGBM, CatBoost + ensembles
DL Models	1D-CNN, 2D-CNN, BiLSTM, CNN-LSTM, Transformer, CrossModal, TabNet
Validation	LOSO (15-fold, subject-independent)
Explainability	TreeSHAP, KernelSHAP, LIME, Grad-CAM, attention maps, clinical reports
Deployment	FastAPI (8 REST + WebSocket), React + TypeScript, Docker

2. Dataset

WESAD (Wearable Stress and Affect Detection) by Schmidt et al. (ICMI 2018). 15 healthy participants completed a protocol with four conditions. Data collected via chest-worn RespiBAN sensor.

Condition	Protocol	Details
Baseline	Neutral reading	20 min. Relaxed magazine reading.
Stress	TSST	5-min public speaking + 5-min mental arithmetic under social evaluation.
Amusement	Funny videos	~7 min of validated humorous clips.
Meditation	Guided breathing	~7 min. Excluded from 3-class classification.

2.1 Sensors

Signal	Rate	Purpose
ECG	700 Hz	Heart rate variability (primary modality)
EDA	700 Hz	Sympathetic nervous system activity
Temperature	700 Hz	Peripheral vasoconstriction during stress
Accelerometer (3-axis)	700 Hz	Body movement and posture
Respiration	700 Hz	Breathing patterns via inductive plethysmography

3. Signal Processing

3.1 ECG

Filter: 4th-order Butterworth bandpass, 0.5–40 Hz.

R-Peak Detection: Pan-Tompkins algorithm with adaptive thresholding.

Artifact Handling: Ectopic beats (>20% deviation from local median RR) removed and replaced via cubic spline interpolation.

Quality Gate: Windows with <40 R-peaks excluded.

3.2 EDA

Filter: 5 Hz lowpass Butterworth.

Decomposition: cvxEDA convex optimization separates tonic (SCL) and phasic (SCR) components.

SCR Detection: Minimum amplitude 0.01 μS. Per-SCR: amplitude, rise time, half-recovery time, AUC.

3.3 Respiration

Filter: Bandpass 0.1–0.5 Hz (6–30 breaths/min).

Detection: Zero-crossing analysis for breath cycles. Outputs: rate, amplitude, variability, inhale/exhale ratio.

3.4 Windowing

60-second windows with 50% overlap. Labels by majority vote from sample-level annotations. Quality-filtered before feature extraction.

4. Feature Engineering

60+ features per window across 7 physiological domains. HRV features follow Task Force of ESC/NASPE standards (Circulation, 1996).

4.1 HRV Time-Domain — 12 features

Feature	Definition
MeanNN	Mean normal-to-normal interval (ms)
SDNN	SD of NN intervals — total variability
RMSSD	Root mean square of successive differences — primary vagal marker
pNN50	% successive differences >50 ms
pNN20	% successive differences >20 ms (more sensitive)
MedianNN	Median NN interval (robust to outliers)
CVNN	SDNN / MeanNN — normalized variability
CVSD	RMSSD / MeanNN
MadNN	Median absolute deviation of NN intervals
MCVNN	MadNN / MedianNN — robust normalized variability
IQRNN	Interquartile range of NN intervals
HRVTI	Total NN count / histogram peak height

4.2 HRV Frequency-Domain — 8 features

Welch PSD: 256-point FFT, Hann window, 50% overlap on 4 Hz interpolated tachogram.

Feature	Band	Significance
VLF Power	0.0033–0.04 Hz	Thermoregulation, renin-angiotensin
LF Power	0.04–0.15 Hz	Mixed sympathetic + parasympathetic
HF Power	0.15–0.4 Hz	Parasympathetic (vagal), respiratory sinus arrhythmia
LF/HF Ratio	—	Sympathovagal balance — elevated during stress
LFn, HFn	—	Normalized powers: LF/(LF+HF), HF/(LF+HF)
Total Power	0.0033–0.4 Hz	Overall autonomic activity
LF Peak Freq	0.04–0.15 Hz	Frequency of maximum LF power

4.3 HRV Nonlinear — 10 features

Feature	Definition
SampEn	Sample Entropy (m=2, r=0.2×SD) — signal complexity, lower in stress
ApEn	Approximate Entropy — includes self-matches
DFA α1	Short-term fractal correlation (4–16 beats)
DFA α2	Long-term fractal correlation (16–64 beats)
SD1	Poincaré plot width — short-term variability (≈RMSSD)
SD2	Poincaré plot length — long-term variability (≈SDNN)
SD1/SD2	Short-to-long-term ratio
CSI	Cardiac Sympathetic Index = SD2/SD1
CVI	Cardiac Vagal Index = log10(SD1 × SD2)
RQA DET	Recurrence Quantification Analysis determinism

4.4 Other Domains

EDA (15): Tonic — SCL mean, std, slope, min, max. Phasic — SCR count, rate, mean/max amplitude, rise time, recovery time, AUC. Statistical — EDA mean, range, kurtosis.

Temperature (5): Mean, std, slope, min, max.

Respiration (5): Rate, amplitude, variability, inhale/exhale ratio, apnea index.

Accelerometer (5): Magnitude, std, zero crossings, energy, peak frequency.

4.5 Signal Image Encoding

For 2D-CNN input, time-series are encoded as images using three methods: GASF (Gramian Angular Summation Field), GADF (Gramian Angular Difference Field), and MTF (Markov Transition Field). Stacked as RGB channels to form 224×224×3 images for pretrained backbones.

5. Statistical Analysis

5.1 Descriptive Statistics

Per-condition summary: mean, std, median, IQR, skewness, kurtosis, 5th/95th percentiles. Normality via Shapiro-Wilk and D'Agostino tests. Outlier detection: IQR-based, z-score, Isolation Forest.

5.2 Hypothesis Testing

ANOVA: Repeated-measures per feature. Sphericity checked via Mauchly's test; Greenhouse-Geisser corrected when violated.

Effect Sizes: η^2, partial η^2, ω^2 (small >0.01, medium >0.06, large >0.14 per Cohen 1988).

Post-hoc: Pairwise tests with Bonferroni, Holm, FDR correction. Pairwise effect sizes: Cohen's d, Hedges' g, Glass's delta.

Non-parametric: Friedman test with Wilcoxon signed-rank post-hoc when normality is violated.

5.3 Dimensionality Reduction

PCA with 95% variance retention. t-SNE (perplexity=30) and UMAP (n_neighbors=15, min_dist=0.1) for 2D visualization. PCA loadings identify top contributing features per component.

5.4 Correlation and Multicollinearity

Pearson, Spearman, Kendall correlation matrices. VIF computed for all features; those with VIF >10 iteratively removed. Feature pairs with $|r|$ >0.8 flagged.

5.5 Mixed-Effects Modeling

Linear mixed-effects: feature ~ condition + (1|subject). Likelihood ratio test, AIC/BIC. ICC quantifies between-subject variance.

5.6 Feature Selection

Combined ranking: ANOVA significance (FDR-corrected $p < 0.05$), effect size ($\eta^2 > 0.06$), and mutual information. Final set balances discriminative power and low multicollinearity.

6. Machine Learning

6.1 Classifiers

Model	Configuration
Logistic Regression	L2, C=1.0, lbfgs, max_iter=1000
SVM	RBF kernel, C=10, gamma=scale, probability=True
Random Forest	200 trees, max_depth=10, min_samples_leaf=5
XGBoost	200 estimators, max_depth=7, lr=0.1, subsample=0.8
LightGBM	50 leaves, 200 estimators, lr=0.1
CatBoost	500 iterations, depth=6, lr=0.1

6.2 Validation

LOSO (primary): 15 folds. Each fold trains on 14 subjects, tests on the held-out subject. Prevents within-subject leakage.

5-fold stratified: For comparison. Typically 5–15% higher than LOSO, showing within-subject overfitting.

Nested CV: Outer LOSO for evaluation, inner 5-fold for hyperparameter selection. Unbiased tuned-model estimates.

6.3 Hyperparameter Optimization

Optuna with TPE sampler. 100 trials per model, 1-hour timeout. Objective: LOSO accuracy. Search spaces: XGBoost (7 params), LightGBM (6), RF (4), SVM (3), CatBoost (4), LR (3).

6.4 Ensembles

Stacking: Top models → out-of-fold probabilities → logistic regression meta-learner.

Voting: Soft (probability averaging) and hard (majority class) with configurable weights.

6.5 Class Imbalance

SMOTE, ADASYN, SMOTE-ENN, class-weighted loss. Resampling applied within each LOSO training fold only.

7. Deep Learning

Architecture	Input	Key Components	Params
1D-CNN	Raw signals	3×Conv1D + BatchNorm + ReLU + GAP, residual connections	~200K
2D-CNN	GASF/GADF/MTF	EfficientNet-B0 pretrained, freeze/unfreeze	~4M
BiLSTM	Feature sequences	2-layer Bidirectional LSTM, hidden=128	~300K
GRU	Feature sequences	Same structure with GRU cells	~200K
CNN-LSTM	Raw signals	CNN encoder → BiLSTM → temporal attention	~500K
Transformer	Feature sequences	4 layers, 8 heads, d_model=128, d_ff=512	~600K
Cross-Modal	Per-modality	Modality encoders + pairwise cross-attention + fusion	~400K
TabNet	Tabular features	5-step attentive selection, n_d=n_a=64	~200K

7.1 Training

Optimizer: AdamW (weight_decay=1e-2).

Scheduler: CosineAnnealingWarmRestarts (T_0=10).

Early Stopping: Patience=15, monitor validation loss, best checkpoint saved.

Mixed Precision: torch.cuda.amp where GPU is available.

Augmentation: Jittering (σ=0.03), scaling (σ=0.1), time warping (σ=0.2), crop (90%), Mixup (α=0.2).

7.2 Cross-Modal Attention

Separate encoder per modality (ECG, EDA, temp, accel). Pairwise cross-attention learns which modalities are complementary for each prediction. Missing modalities handled via masking. Modality importance scores extracted for interpretability.

8. Explainability

8.1 SHAP

TreeSHAP for tree-based models (exact Shapley values). KernelSHAP for SVM and neural networks. Visualizations: beeswarm summary, waterfall, force plot, dependence plots with interaction coloring.

8.2 LIME

Local linear surrogate per prediction. Top-k features identified. Rankings cross-checked against SHAP for consistency.

8.3 Attention and Grad-CAM

Temporal attention weights from CNN-LSTM/Transformer overlaid on input signals. Cross-modal attention weights show modality importance. Grad-CAM highlights relevant regions of GASF/GADF/MTF images for 2D-CNN.

8.4 Clinical Interpretation

Feature values mapped to published reference ranges (Task Force 1996 for HRV; published EDA norms). RMSSD normal: 25–45 ms. SDNN: 50–100 ms. LF/HF: 1.0–2.0. SCL: 2–16 μS. Each prediction generates a report with biomarker status, top SHAP contributors, physiological narrative, and stress severity (none / mild / moderate / high).

9. Deployment

9.1 API

Method	Endpoint	Description
POST	`/api/predict`	Single prediction with probabilities and SHAP explanation
POST	`/api/predict-batch`	Batch inference for multiple samples
POST	`/api/process-signal`	Raw CSV → preprocessing → features → prediction
GET	`/api/models`	List available models with metrics
POST	`/api/explain/shap`	SHAP values and visualization
POST	`/api/explain/lime`	LIME local explanation
POST	`/api/explain/clinical`	Clinical biomarker interpretation report
WS	`/ws/stream`	Real-time streaming predictions via WebSocket

9.2 Dashboard

React 18 + TypeScript + Tailwind CSS. Five views: Dashboard (metrics overview), Signal Explorer (multi-track Plotly visualization with R-peak/SCR markers), Prediction Panel (CSV upload, confidence gauge), Explainability (SHAP/LIME charts, clinical report card), Model Comparison (metrics table, confusion matrices, ROC curves, per-subject heatmap).

9.3 Infrastructure

Multi-stage Docker: Node 18 frontend build → Python 3.11-slim runtime. Docker Compose with volume mounts. GitHub Actions CI runs syntax check and test suite on every push.

10. Testing

Test File	Tests	Scope
test_preprocessing.py	44	ECG, EDA, respiratory processors, windowing, pipeline
test_features.py	53	7 extractors, image encoder, extraction pipeline
test_analysis.py	27	ANOVA, PCA, correlation, mixed effects, feature selection
test_ml_models.py	56	6 classifiers, LOSO/KFold, Optuna, ensembles, evaluation
test_dl_models.py	35	7 architectures, data loaders, trainer, augmentation
test_explainability.py	~35	SHAP, LIME, clinical interpreter, attention, reports
test_api.py	~36	All endpoints, WebSocket, model manager, schemas
Total	**286**	**Full pipeline coverage**

10.1 Code Standards

Type hints on all functions. Google-style docstrings. Centralized config (no hardcoded values). Logging via LoggerMixin (no print statements). pathlib.Path for all file operations. Specific exception handling with meaningful messages.

11. Project Structure

Directory	Files	Contents
`src/preprocessing/`	6	ECG, EDA, respiratory processors, windowing, pipeline, base class
`src/features/`	9	HRV (time, freq, nonlinear), EDA, temp, resp, accel, image encoder, pipeline
`src/analysis/`	6	Descriptive, hypothesis, dim reduction, correlation, mixed effects, selection
`src/models/ml/`	8	Base, 6 classifiers, CV, tuning, ensemble, evaluation, imbalance, pipeline
`src/models/dl/`	12	7 architectures, trainer, augmentation, data loader, registry, base
`src/explainability/`	5	SHAP, LIME, Grad-CAM, attention visualizer, clinical interpreter
`api/`	7	Main, schemas, model manager, routes (prediction, models, explanation, WS)
`frontend/src/`	~10	Dashboard, SignalExplorer, PredictionPanel, Explainability, ModelComparison
`tests/`	7	One test file per module
`notebooks/`	8	01-Data, 02-Signals, 03-Features, 04-Stats, 05-ML, 06-DL, 07-XAI, 08-Deploy

12. Complete Techniques List

Signal Processing (9)

Butterworth bandpass/lowpass filtering, Pan-Tompkins R-peak detection, cvxEDA decomposition, cubic spline interpolation, ectopic beat removal, SNR quality estimation, zero-crossing breath detection, windowing with majority-vote labeling.

Feature Engineering (14)

HRV time-domain (12 metrics), Welch PSD frequency-domain (8 metrics), Sample/Approximate Entropy, DFA short/long-term, Poincaré analysis (SD1/SD2/CSI/CVI), RQA determinism, EDA tonic/phasic features (15 metrics), SCR characterization, GASF, GADF, MTF, RGB stacking.

Statistical Analysis (20)

Shapiro-Wilk, D'Agostino normality, IQR/z-score/Isolation Forest outlier detection, repeated-measures ANOVA, Mauchly sphericity, Greenhouse-Geisser correction, Bonferroni/Holm/FDR correction, Cohen's d/Hedges' g/Glass's delta, η^2/ω^2, Friedman, Wilcoxon signed-rank, PCA, t-SNE, UMAP, Pearson/Spearman/Kendall correlation, VIF, mutual information, LMM, ICC.

Machine Learning (12)

Logistic Regression, SVM, Random Forest, XGBoost, LightGBM, CatBoost, stacking ensemble, voting ensemble, Optuna TPE, nested CV, LOSO CV, SMOTE/ADASYN/SMOTE-ENN/class weighting.

Deep Learning (15)

1D-CNN (residual), EfficientNet-B0 transfer, BiLSTM, GRU, CNN-LSTM+attention, Transformer (learnable/sinusoidal positional encoding), cross-modal attention, TabNet, AdamW, cosine annealing, early stopping, mixed precision, jittering/scaling/warping, Mixup, random crop.

Evaluation (10)

Accuracy, balanced accuracy, F1 (macro/weighted), precision, recall, MCC, AUC-ROC, AUC-PR, confusion matrix, McNemar, Wilcoxon model comparison, paired t-test, bootstrap CI, per-subject analysis.

Explainability (8)

TreeSHAP, KernelSHAP, SHAP dependence/interaction, LIME, Grad-CAM, temporal attention extraction, cross-modal attention visualization, clinical biomarker interpretation.

13. References

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